

```

export default function StateExampleComponent() {
  let [name, setName] = useState("Sunny");
  let [age, setAge] = useState(18);

  return (
    <div>
      Name: {name} <br />
      Age: {age} <br />
      <button type="button"
        onClick={() => setAge(age+1)}>Increment Age</button>
      <button type="button"
        onClick={() => setName(faker.person.fullName())}>Change Name</button>
    </div>
  );
}

```

```

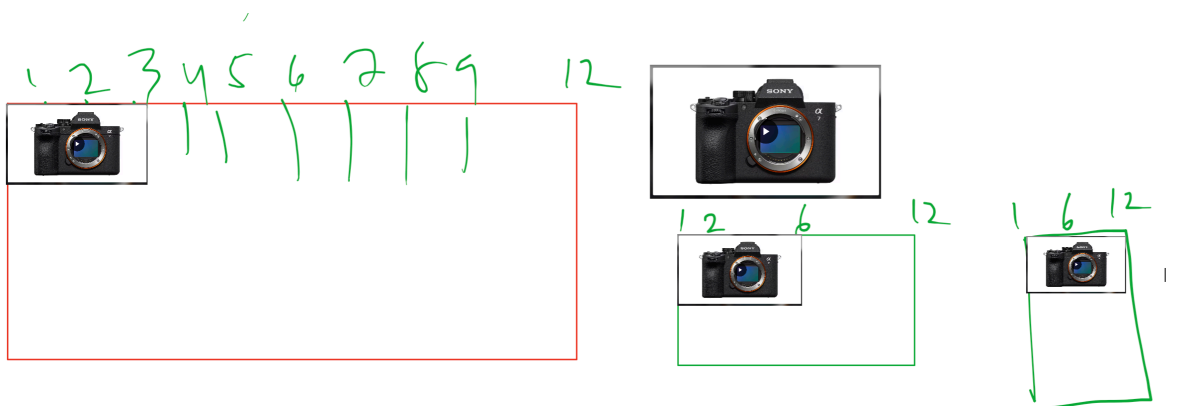
export default class StateExampleComponent extends Component {
  state = {
    name: "Sunny",
    age: 18
  };

  updateName(name) {
    this.setState({
      name
    });
  }

  updateAge(age) {
    this.setState({
      age
    });
  }
}

```

~~age = age + 1~~
 ↳ ~~Reconciliation~~



``

```

function memo(fn) {
  var cache = {};
  return function(args) {
    if(!cache[args]) {
      cache[args] = fn(args);
    }
    return cache[args];
  }
}

```

40

```

function fibanocci(no) {
  if(no == 0 || no == 1) {
    return 1;
  } else {
    return fibanocci(no - 1) + fibanocci(no - 2);
  }
}

```

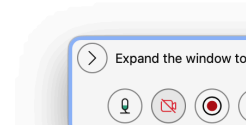
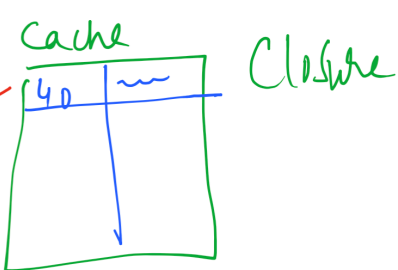
`let memFib = memo(fibanocci);`

```

console.time("first");
1 console.log(memFib(40));
console.timeEnd("first");

console.time("second");
2 console.log(memFib(40));
console.timeEnd("second");

```



```
export default function countReducer(state, action) {
  switch(action.type) {
    case 'INCREMENT':
      return {
        count: state.count + action.payload
      }
    case 'DECREMENT':
      return {
        count: state.count - 1
      }
    case 'RESET':
      return {
        count: 0
      }
    default:
      return state
  }
}
```

```
let initialState = {
  count: 0
}
export default function ReducerHookComponent() {
  let [state, dispatch] = useReducer(countReducer, initialState);
  return (
    <div>
      Count { state.count} <br />
      <button type="button"
        onClick={() => dispatch({type: 'INCREMENT', payload: 10})}>
        Increment
      </button>
      <button type="button"
        onClick={() => dispatch({type: 'DECREMENT'})}>
        DECREMENT
      </button>
      <button type="button"
        onClick={() => dispatch({type: 'RESET'})}>
        RESET
      </button>
    </div>
  )
}
```

```
{
  count: 0
}
```

```
export default function countReducer(state, action) {
  switch(action.type) {
    case 'INCREMENT':
      return {
        count: state.count + action.payload
      }
    case 'DECREMENT':
      return {
        count: state.count - 1
      }
    case 'RESET':
      return {
        count: 0
      }
    default:
      return state
  }
}
```

```
let initialState = {
  count: 0
}
export default function ReducerHookComponent() {
  let [state, dispatch] = useReducer(countReducer, initialState);
  return (
    <div>
      Count { state.count} <br />
      <button type="button"
        onClick={() => dispatch({type: 'INCREMENT', payload: 10})}>
        Increment
      </button>
      <button type="button"
        onClick={() => dispatch({type: 'DECREMENT'})}>
        DECREMENT
      </button>
      <button type="button"
        onClick={() => dispatch({type: 'RESET'})}>
        RESET
      </button>
    </div>
  )
}
```

```
{
  count: 10
}
```

```
export default function countReducer(state, action) {
  switch(action.type) {
    case 'INCREMENT':
      return {
        count: state.count + action.payload
      }
    case 'DECREMENT':
      return {
        count: state.count - 1
      }
    case 'RESET':
      return {
        count: 0
      }
    default:
      return state
  }
}
```

```
let initialState = {
  count: 0
}
export default function ReducerHookComponent() {
  let [state, dispatch] = useReducer(countReducer, initialState);
  return (
    <div>
      Count { state.count} <br />
      <button type="button"
        onClick={() => dispatch({type: 'INCREMENT', payload: 10})}>
        Increment
      </button>
      <button type="button"
        onClick={() => dispatch({type: 'DECREMENT'})}>
        DECREMENT
      </button>
      <button type="button"
        onClick={() => dispatch({type: 'RESET'})}>
        RESET
      </button>
    </div>
  )
}
```

```
{
  count: 9
}
```